Vector Magnitude & Direction: Cartesian to Polar

Right Triangle Trigonometry & Vectors • Standard: CCSS.MATH.HSN.VM.A.1

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1. MATHEMATICAL CONCEPT & STACK APPROACH: Decompose 2D vectors into horizontal and vertical components on the stack (Y = Δx, X = Δy) and convert to polar coordinates (r, ϑ).

Every 2D vector or right triangle relates two perpendicular legs (Δx, Δy) to a hypotenuse r and angle ϑ via $r = \sqrt{(x^2 + y^2)}$ and $\tan(\vartheta) = y/x$. By placing Δy in Register Y and Δx in Register X, the stack keeps both components paired to compute magnitude and angle simultaneously.

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Right-Triangle & 2D Vector Component Diagram

Hands-On Setup: Draw a right triangle with horizontal run Δx = 4 (Register Y) and vertical rise Δy = 3 (Register X) to model the 3-4-5 vector.

Student Prediction (Before Calculator): Run = 4, Rise = 3. Predict magnitude: r = sqrt($4^2 + 3^2$) = 5. Predict angle: arctan(3/4) ~ 36.87°.

3. CALCULATOR EXECUTION (SC-32 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: [4] [ENTER] [3] [↶] [▸θ,r] [x⇄y]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[4]	4	Enter horizontal component x = 4
[ENTER]	4	4 lifted to register Y
[3]	3	Enter vertical component y = 3 in X
[↶] [▸θ,r]	36.87	Rectangular to Polar! Screen displays angle ϑ = 36.87°
[x⇄y]	5.00	Swap registers: Y displays vector magnitude r = 5.00!

4. STORY CONTEXT (A Stick in Alexandria: How Eratosthenes Measured the Earth with a Shadow & Crosswind Vectors)

In 240 BC, Eratosthenes measured a 7.2° shadow angle in Alexandria at noon on the summer solstice while a well in Syene 800 km away had zero shadow—calculating Earth's circumference within 1% using pure geometry! Using 84.0 mm Fraction Studio ratio towers and [→ϑ,r] polar conversion, students resolve right triangles and aircraft crosswind vectors in a single keystroke.

Core Investigation (Student Warm-Up A):

A surveyor walks 60 m East and 80 m North. Find the straight-line displacement magnitude and heading.

2D Vector Addition: Resultant Forces

Vector Addition & Resultant Forces • Standard: CCSS.MATH.HSN.VM.A.3

1. EXTENSION CONCEPT (WARM-UP B), BLOCKS & PREDICTION

Concept & Stack Approach: Non-collinear vectors cannot be added by simply summing their lengths; their horizontal (x) and vertical (y) components must be summed separately first. The 4-level stack accumulates ☒x and ☒y from the inside out before converting the resultant back to polar form.

Blocks & Prediction: Sketch component vectors F1 = <3, 2> and F2 = <1, 5> tip-to-tail on a coordinate grid. Prediction: Add components: x_net = 3 + 1 = 4; y_net = 2 + 5 = 7. Magnitude r = sqrt($4^2 + 7^2$) = sqrt(65) ~ 8.06.

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[3] [ENTER] [1] [+] [2] [ENTER] [5] [+] [↰] [▸θ,r] [x⇄y]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[3]	3	Horizontal component x1 = 3
[ENTER]	3	Lifted to Y
[1]	1	Horizontal component x2 = 1
[+]	4	Net horizontal force x_net = 4 in X
[2]	2	Vertical component y1 = 2 (4 lifts to Y)
[ENTER]	2	2 lifted to Y; 4 in Z
[5]	5	Vertical component y2 = 5 in X
[+]	7	Net vertical force y_net = 7 in X; x_net = 4 in Y
[↰] [▸θ,r]	60.26	Resultant angle ϑ = 60.26°
[x⇄y]	8.06	Resultant force magnitude r = 8.06!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Evaluating trigonometric functions in Radian mode when angles are given in Degrees (or vice versa).

Hands-On Block Remediation: Have students check the DEG/RAD indicator on the SC-32 status bar and verify with the 1/2 (42 mm) and 1 Whole (84 mm) Fraction Studio towers that sin(30°) = 42/84 = 0.5000 before starting vector problems.

Misconception 2: Swapping x and y components when converting rectangular coordinates to polar via [→ϑ,r].

Hands-On Block Remediation: Place horizontal Run (x) in Row 1 (Register Y) and vertical Rise (y) in Row 0 (Register X) of the Place Value Tray before pressing [→ϑ,r].

4. EXTENSION MATHEMATICAL INSIGHT

Non-collinear vectors cannot be added by simply summing their lengths; their horizontal (x) and vertical (y) components must be summed separately first. The 4-level stack accumulates ☒x and ☒y from the inside out before converting the resultant back to polar form.

Trigonometry, Vectors & Polar/Rectangular Coordinates

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: VECTOR DECOMPOSITION & POLAR CONVERSION RULES

- Trigonometric functions are ratios of right-triangle sides: sin(ϑ) = Opposite / Hypotenuse, cos(ϑ) = Adjacent / Hypotenuse, tan(ϑ) = Opposite / Adjacent.
- A 2D vector v = <x, y> places horizontal run Δx in Register Y (Waiting Shelf) and vertical rise Δy in Register X (Workbench).
- Rectangular to Polar [f:] [→ϑ,r]: Converts (x, y) into angle ϑ (in X) and magnitude r = √(x² + y²) (in Y).

Key Sequence: `4` `ENTER` `3` `↶` `▸ϑ,r` `x ≷ y`

Investigation 1: Find magnitude and angle for vector <12, 5>.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Stack Plan:

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 2: Find the hypotenuse of a right triangle with legs 7 and 24.

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Stack Plan:

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 3: Compare the 1/2 tower (42 mm) to the 1 Whole tower (84 mm) and calculate sin(30°).

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Stack Plan:

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 4: A vector has magnitude 10 at angle 45°. What are its x and y components?

1. Geometric / Area / Coordinate Sketch:

2. Algebraic Setup & Stack Plan:

Algebraic Setup:

Stack Registers (Y, X):

RPN Keystroke Plan:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Unit 09: Trigonometry, Vectors & Polar/Rectangular Coordinates — Answer Key

Complete solutions and SC-32 verification keystrokes for Grades 9–11

1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): A surveyor walks 60 m East and 80 m North. Find the straight-line displacement magnitude and heading.

Core Solution: Displacement: 100 m, Heading: 53.13°. Keystrokes: [6][0] [ENTER] [8][0] [f:][→ϑ,r] [x swap y]

Core Check Keys: 6 0 ENTER 8 0 ↶ ▸θ,r x⇄y

Extension Challenge (Warm-Up B): Two cables pull on a post: Vector A is <5, 2> and Vector B is <-2, 4>. Find the resultant vector magnitude.

Extension Solution: Magnitude: 6.71. Resultant: <3, 6>.

Ext. Check Keys: 5 ENTER 2 +/− + 2 ENTER 4 + ↶ ▸θ,r x⇄y

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Find magnitude and angle for vector <12, 5>.

Expected Answer: ϑ = 22.62°, r = 13

SC-32 Verification Keys: 1 2 ENTER 5 ↶ ▸θ,r

Investigation 2: Find the hypotenuse of a right triangle with legs 7 and 24.

Expected Answer: 25

SC-32 Verification Keys: 7 ENTER 2 4 ↶ ▸θ,r x⇄y

Investigation 3: Compare the 1/2 tower (42 mm) to the 1 Whole tower (84 mm) and calculate sin(30°).

Expected Answer: 0.5

SC-32 Verification Keys: 3 0 SIN

Investigation 4: A vector has magnitude 10 at angle 45°. What are its x and y components?

Expected Answer: x = 7.07, y = 7.07

SC-32 Verification Keys: 1 0 ENTER 4 5 ↷ ▸y,x

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.